a **Validation: distinct denominators**

Comparison / agreement criterion	Met / tested units	Not met, n
8088/hardware: identical vector states	3,126,917/3,137,000	10,083
Corrected/historical: below change-rate bound	643/690	47
Corrected/historical: identical assay metrics	407/417	10
Corrected/historical: unchanged claim verdicts	19/22	3
Corrected/historical: unchanged components	32/36	4

All 10,083 hardware divergences were expected

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
100,000 encounter draws/condition; units named in each row

b **Stability flags and assay differences**

Core	Flagged / conditions	Assays differing / tested
Corrected	47/690	10/417
C2	114/690	10/417

Comparator: historical compatibility core C0

Corrected core (C1): classic; core C2: Zilog NMOS

Overall stability verdict: criterion not met for either core

Stable: upper 95% CI of changed-final-memory pair fraction <0.01%
Flag = stability criterion not met; assays compare with core C0

c **I1-T2: all 22 claim verdicts**

Claim	vs archive	Claim	vs archive
DZ1	stands	C4b	stands
DZ2	**changed**	C4c	stands
DZ3	**changed**	F4c	stands
E1	stands	S1	stands
E2	stands	S2	stands
E3	stands	S3	stands
E-L32L	stands	S4	stands
L1	stands	S5	stands
L2	**changed**	X1-1	stands
L3	stands	X1-2	stands
L	stands	X1-3	stands

DZ2 timing; DZ3 transmitter trend; L2 recovery sites
Stands = same as archive; E1 stability remains failed; 19/22 stand

d **Four changed components of 36**

Component	Historical	Corrected core
DZ2	not met	met
DZ3	not met	met
L2 L16	not met	n/a
L2 L20	n/a	not met

n/a = not applicable: no recovered lineage

Component tests summarized in panel a; L = tape bytes

DZ2/DZ3: criteria now met; L2: applicability changes

Historical = published archive; DZ timing/share; L2 sites
Complete 36-component ledger retained in Source Data

e **Assay changes by measurement**

Metric	Differing corrected core	Differing C2
Partner-copy fraction, parent A	0	0
Partner-copy fraction, parent B	6	6
Initial byte-match fraction	0	0
Informative partners, generation 1 (n)	0	0
Informative partners, generation 2 (n)	0	0
Cyclic copy offset (bytes)	0	0
Parent-byte fraction preserved, A	0	0
Parent-byte fraction preserved, B	9	9
Offspring ≥75% parent-match fraction	0	0
Generation-2 normalized gain	0	0
Generation-2 conditional gain	0	0
Normalized culture gain	0	0
Generation-2 gain ≥0.3 (yes/no)	0	0
Faithful culture classification (yes/no)	0	0

Counts /417 assays vs core C0; rows overlap (not additive)
A/B = parent role; copy = ≥75% immediate-parent byte match

f **Stage-L scoring and snapshot availability**

L bytes	Worlds, n	Scoreable from final data, n	Missing 7,500-update snapshot, worlds
16	10	10	10
20	10	10	10

All worlds completed 10,000,000 updates

Retained final bytes and journals support scoring

Applied-mutation/collision counts unavailable for these worlds

Stage L: closure-recovery worlds, 10 / tape length
The missing observation is a world snapshot at 7,500 updates